

Code-Layout, Readability and Reusability

Date	04 July 2026
Team ID	
Project Name	A Comprehensive Measure of Well-Being
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S. No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout The code.	Yes	Proper indentation is maintained throughout the code.
2	Proper File Structure	Files and folders are logically organized	Yes	Project files are organized into Dataset, Templates, Static, Model, and App files.
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Variables such as df, X, y, and model clearly represent their purpose.
4	Function / Method Names	Functions are descriptively named	Yes	Functions are named clearly and describe their functionality.
5	Code Comments	Inline and block comments explain log	Partial	Basic comments are included to explain important sections.
6	Modular Design	Code is split into reusable functions/m	Yes	Code is divided into data preprocessing, model training, and web application modules.
7	No Redundant Code	Duplicate or unused code is removed	Yes	Unused and duplicate code has been removed.

8	Error Handling	Exceptions and errors are handled gracefully	Partial	Basic error handling is implemented for user input and file loading.
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Reusable Components / Modules:

S. No	Component/Module Name	Language/Technology	Where Reused	Reusability Level (High/Medium/Low)
1.	Data Preprocessing	Python, pandas	Model Training	High
2.	Machine Learning Model	Scikit-learn	Prediction Module	High
3.	Flask Web Application	Flask, python	User Interface	High
4.	HTML Templates	HTML, CSS	Web Pages	Medium
5.	Model File (HDI. pickle)	Pickle	Prediction System	High

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Project is well organized with separate folders and files.
Readability	5	Code is easy to understand with meaningful variable names.
Reusability	5	Modules can be reused in similar machine learning projects.
Documentation / Comments	4	Basic comments are provided; additional comments can improve clarity.
Overall Score	4.8	Well-Structured, readable, and reusable machine learning project.